

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech (IT) Examination

March-April 2018

IT710 Soft Computing

Date: 03/04/2018, Tuesday

Time: 10.00 a.m. To 1.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below. [07]

1. In sigmoidal function, the graph is _____. **01**
2. Networks which have feedback connection are called _____. **01**
3. Minimum _____ numbers of layers are required for multi-layer perceptron. **01**
4. Adaptive filters are used for _____ type of learning. **01**
5. What do you mean by soft computing? List down the technique of soft computing. **03**

Q – 2.a How soft computing is better than the conventional computing? [04]

Q – 2.b Answer any two questions. [10]

1. Give the problem statement of travelling salesman problem. Solve the same problem using Genetic Algorithm.
2. Explain Roulette Wheel Selection With Example.
3. Define Following Terms with Respect to Genetic Algorithm.
(i) Reproduction (ii) Crossover.

Q - 3 Answer any two questions. [14]

1. What is Genetic algorithm? How it is different from traditional algorithm?
2. Explain how Genetic algorithm used for weight optimization in neural network.
3. What do you mean by GA. Discuss the different designing Issues for Genetic Algorithms?

SECTION – II

- Q - 4 Answer the question below. [07]**
1. Traditional set theory is also known as _____. **01**
 2. State True or False **02**
 - a) Probability density function means probability distribution for continuous variable.
 - b) The values of the set membership is represented by probabilities.
 3. What is sequence of steps taken in designing a fuzzy logic machine? **01**
 4. What is the Importance of linguistic variable in fuzzy theory? **01**
 5. What is an adaptive fuzzy system? **02**
- Q – 5.a What do you mean by artificial Neural Network? List down the features that make Neural Network more powerful. [05]**
- OR**
- Q – 5.a What do you mean by Neural Network. Draw the simple architecture of neural network. What are the significances of Hidden layer in Neural network? [05]**
- Q – 5.b Draw the architecture of simply recurrent network and fully recurrent network. [04]**
- Q – 5.c What is the use of activation function in neural network? Explain step function and sign function in neural network. [05]**
- OR**
- Q – 5.c Explain back propagation training algorithm with suitable example. [05]**
- Q – 6.a Explain following terms with respect to NN [05]**
- (i) Linear separability (ii) Supervised learning.
- OR**
- Q – 6.a What do you mean by Fuzzy Inference System? Explain Mamdani Fuzzy Inference system with suitable example. [05]**
- Q – 6.b Draw and explain the Architecture of fuzzy logic controller. [04]**
- Q – 6.c Explain following defuzzification method in with suitable example. [05]**
- (i) Center of Sums Method (COS) (ii) Center of Area / Bisector of Area Method (BOA)
- OR**
- Q – 6.c Explain Hybrid Fuzzy Genetic Algorithm with suitable example. [05]**
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